RADIONETTE — COMPLETE WIRING OVERVIEW

Raspberry Pi 4 • BCM numbering • 3.3 V logic

INPUT CONTROLS

Contact / switch	Pi input	Physical
Channel bit 0	BCM 18	pin 12
Channel bit 1	BCM 23	pin 16
Channel bit 2	BCM 24	pin 18
Channel bit 3	BCM 25	pin 22
Channel bit 4	BCM 5	pin 29
Channel bit 5	BCM 7	pin 26
Channel bit 6	BCM 12	pin 32
Channel bit 7	BCM 16	pin 36
Bluetooth mode	BCM 20	pin 38
Power sense	BCM 21	pin 40
Stereo select	BCM 19	pin 35

Each active contact connects its GPIO to 3.3 V. Software enables an internal pull-down. LOW = inactive; stereo input LOW selects mono.

SHARED I²C BUS 1

ADS1115 — address 0x48

VDD → 3V3 (pin 1 or 17)
GND → Pi GND
SDA → BCM 2 / pin 3
SCL → BCM 3 / pin 5
ADDR → GND
AIN0 → volume pot wiper
AIN1–3 → not connected

Pot outer pins: 3V3 ↔ GND

AS5600 — address 0x36

VDD → shared 3V3
GND → shared GND
SDA → BCM 2 / pin 3
SCL → BCM 3 / pin 5
DIR → GND
OUT → not connected
PGO → not connected

Diametric magnet 0.5–3 mm above sensor on tuning shaft.

11 × GPIO

SDA

SCL

RASPBERRY PI 4 — 40 PIN HEADER

BUS / DISPLAY

3V3 pin 1 / 17
GND any GND pin
SDA1 BCM 2 pin 3
SCL1 BCM 3 pin 5
SPI CE0 BCM 8 pin 24
SPI MOSI BCM 10 pin 19
SPI SCLK BCM 11 pin 23
Display DC BCM 27 pin 13
Display RST BCM 22 pin 15
Backlight BCM 13 pin 33

Audio: 3.5 mm / HDMI

Network: Wi-Fi + BT

All logic is 3.3 V. Never apply 5 V to GPIO, I²C, SPI, or control inputs.

GPIO CONTROL

CH0 BCM18 pin12
CH1 BCM23 pin16
CH2 BCM24 pin18
CH3 BCM25 pin22
CH4 BCM5 pin29
CH5 BCM7 pin26
CH6 BCM12 pin32
CH7 BCM16 pin36
BT BCM20 pin38
PWR BCM21 pin40
MODE BCM19 pin35

12V RELAY BCM17 pin11
BT LED BCM26 pin37

SPI

GPIO

GC9A01 ROUND DISPLAY — SPI0

Signal	Pi	Function
CS	BCM 8 / 24	SPI CE0
SCL	BCM 11 / 23	SPI clock
SDA	BCM 10 / 19	SPI MOSI
DC	BCM 27 / 13	data / command
RES	BCM 22 / 15	reset
Backlight	BCM 13 / 33	active HIGH
VCC	3V3 / pin 1	2.8–3.3 V
GND	GND	common

Use either display variant shown on page 2 — not both.

RELAY + INDICATOR OUTPUTS

GPIO 17 / pin 11 → relay module IN (active HIGH)

GPIO 26 / pin 37 → resistor* → Bluetooth LED → GND

Relay control side is powered by the Pi; contacts switch the separate 12 V supply.

POWER + AUDIO

Pi supply + GND → relay module
GPIO 17 → IN (active HIGH)
Relay COM ← separate 12 V +
Relay NO → amp + radio lights

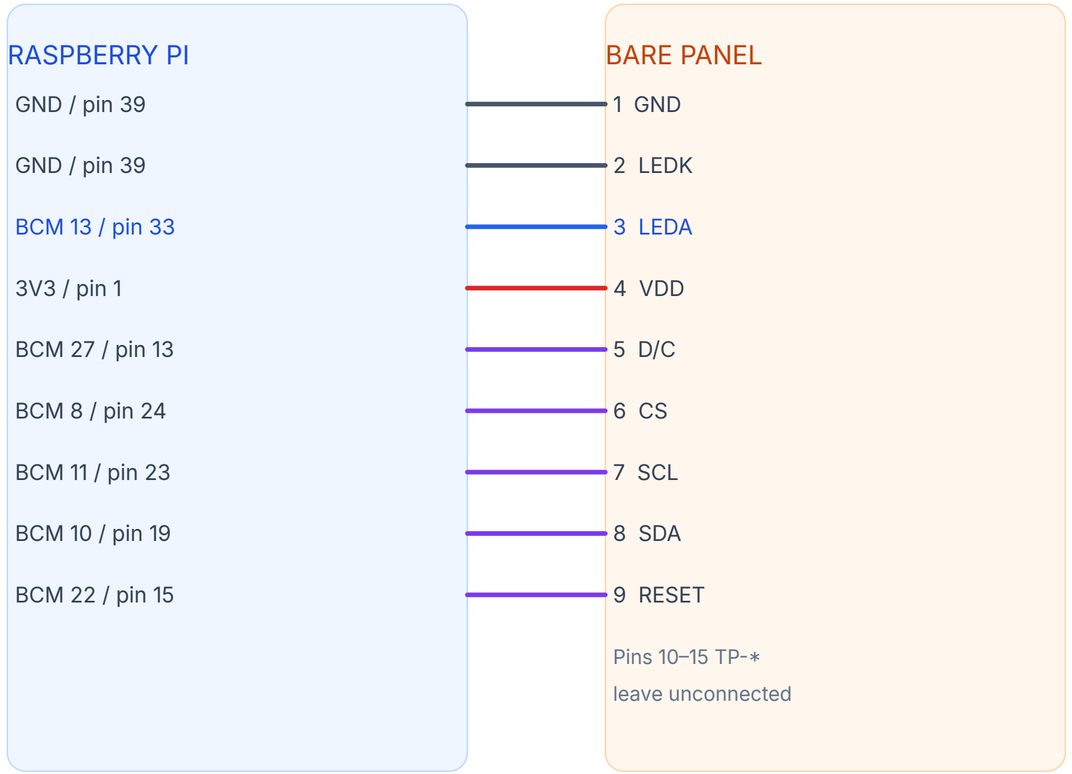
Switched 12 V from relay NO
→ amplifier + radio lights
Pi audio output → amplifier input
12 V − → amp/lights return

RADIONETTE — GC9A01 DISPLAY VARIANTS

Existing bare panel and optional 8-pin PCB module

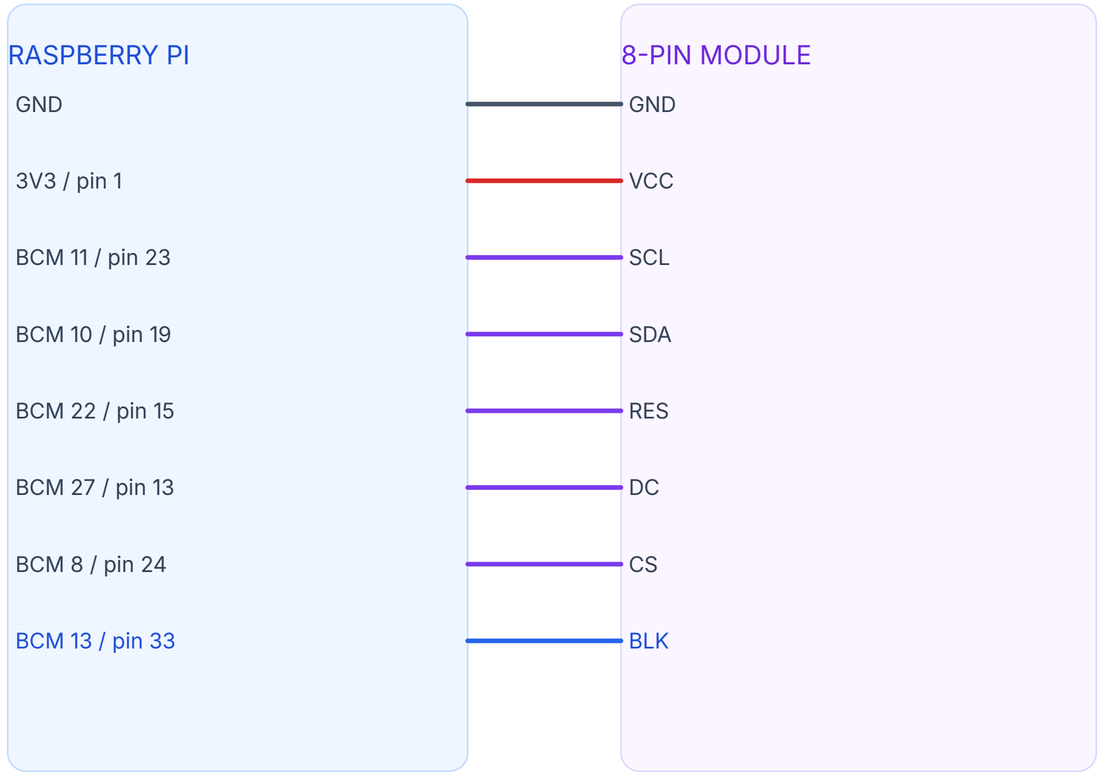
A — EXISTING BARE 14/15-PIN PANEL

As built: backlight connected directly; no MOSFET



B — OPTIONAL 8-PIN PCB MODULE

GC9A01 • 240×240 • 2.8–3.3 V • 4-wire SPI



INSTALLATION NOTES

- Power the Pi off before changing display wiring.
- Both variants use the same SPI, DC and reset signals.
- GPIO 13 is active HIGH: on while HIGH, off while LOW.
- Software auto-off is 20 seconds after the last visible event.
- Use 3.3 V only; the new module is specified for 2.8–3.3 V.
- Existing bare panel is documented exactly as built: LEDA directly to GPIO 13 and LEDK to GND.
- Before duplicating the bare-panel wiring, verify backlight current is within GPIO limits.